

Dense sets with no nontrivial sum-product identity

Counterexample to Conjecture 00000000038

Abstract

For every integer $N \geq 4$, the set $A_N = \{3, 4, \dots, N\}$ has density at least $1/2$ in $[N]$ and admits no identity $x_1 + \dots + x_k = x_1 \cdots x_k$ of length $k \geq 2$, even with repeated choices. Thus Conjecture 00000000038 is false. The accompanying Lean 4 proof verifies the inequality for arbitrary finite lists, the exact cardinality and density, arbitrarily large counterexamples, and the negation of the conjecture even with an additional eventual- N threshold.

1 Statement and the nontriviality requirement

The source asserts that for every $\delta > 0$ there is a length $k(\delta)$ such that every $A \subseteq [N]$ with density at least δ contains a nontrivial choice of k elements, allowing repetition, whose sum equals their product. Here $[N] = \{1, \dots, N\}$ and density means $|A|/N$.

Both the English and Chinese source explicitly require nontriviality. A one-term list always satisfies $x_1 = x_1$, independently of A , so it is the trivial case that must be excluded. We prove the stronger and unambiguous fact that *every* sum-product identity in the proposed counterexample has length exactly one. Hence the counterexample applies to any nontriviality convention excluding these singleton identities. If singleton identities were admitted, taking $k = 1$ would instead make the assertion immediate and render the source's nontriviality clause meaningless. Repetition is fully permitted in the proof below.

2 An elementary obstruction

Lemma 1. *For every integer $k \geq 2$ and every sequence of integers $x_1, \dots, x_k \geq 3$,*

$$\sum_{i=1}^k x_i < \prod_{i=1}^k x_i.$$

Proof. If $u, v \geq 3$, then

$$uv - u - v = (u - 1)(v - 1) - 1 \geq 2 \cdot 2 - 1 = 3 > 0.$$

This proves the case $k = 2$. Suppose now that $k \geq 3$ and the result holds for $k - 1$ terms. Put

$$a = x_1, \quad S = \sum_{i=2}^k x_i, \quad P = \prod_{i=2}^k x_i.$$

Then $a \geq 3$, $S \geq 6$, and $S < P$. Applying the two-variable inequality to a, S gives

$$a + S < aS < aP,$$

which completes the induction. No distinctness of the x_i was used. $\square$

Theorem 1. *For every $N \geq 4$, let $A_N = \{3, 4, \dots, N\}$. Then*

$$A_N \subseteq [N], \quad |A_N| = N - 2, \quad \frac{|A_N|}{N} = 1 - \frac{2}{N} \geq \frac{1}{2}.$$

For every $k \geq 2$ and every choice $x_1, \dots, x_k \in A_N$, with repetitions allowed, the sum differs from the product.

Proof. The inclusion and cardinality follow directly from the endpoints of the integer interval. Since $N \geq 4$, $N - 2 \geq N/2$, proving the density claim. Every selected element is at least 3, so Lemma 1 gives a strict inequality and in particular excludes equality. $\square$

Fix $\delta = 1/2$. Any proposed nontrivial length $k(1/2)$ is excluded by Theorem 1. Moreover, the counterexamples are arbitrarily large: for any proposed bound B , choose $N \geq \max\{B, 4\}$. Thus inserting a “for all sufficiently large N ” qualification would not repair the conjecture. Nor would allowing the chosen length to depend on A or N , because A_N excludes all lengths $k \geq 2$ simultaneously.

3 Formal verification and scope

The project pins Lean v4.19.0 and Mathlib commit c44e0c8ee63ca166450922a373c7409c5d26b00b.

The file `lean/Main.lean` contains the following kernel-checked statements:

- `sum_lt_prod`
The strict inequality for an arbitrary finite list with all entries at least 3 and length at least 2.
- `only_singleton_solutions`
Any such list whose sum equals its product has length exactly one.
- `counterexampleSet_half_dense`
The real-valued density inequality, proved from the separately checked exact interval cardinality.
- `arbitrarily_large_dense_counterexamples`
A counterexample above every bound B on the interval size.
- `no_uniform_length_even_eventually`
The negation of the fixed-density assertion, even after allowing a threshold N_0 .

In the formalization, A_N is the standard finite integer interval `Finset.Icc 3 N`, and density is the actual real quotient `(A.card : Real) / (N : Real)`. Selections are ordinary finite lists of natural numbers, so repeated entries are retained rather than removed. The final theorem has the logical shape

$$\neg \exists k \geq 2 \exists N_0 \forall N \geq N_0, N > 0 \implies \left[\begin{array}{l} \text{every } A \subseteq [N] \text{ with } |A|/N \geq 1/2 \\ \text{has a length-}k \text{ sum-product identity} \end{array} \right].$$

This is a negation of a weaker assertion than the source’s universal claim, so it suffices to refute that claim. The only semantic point is the source’s explicit exclusion of trivial singleton identities; no replacement arithmetic, density, or solution relation is used.

The full infinite family is proved by induction and exact algebra. There is no finite search or numerical experiment standing in for a universal proof. The Lean file contains no unfinished proofs or added axioms. The printed axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`.

Source. The Last Math Competition, Conjecture 00000000038. The exact bilingual statement is preserved byte for byte in `SOURCE.md`. Build instructions are in `README.md`; source provenance and self-review are in the `verification` directory.